

Review Paper: Commercial Haptic Gloves

The point of this paper is that there are many unique haptic glove designs and that all of them have potential. However, due to the sheer quantity of designs it is “clear that not all new ideas will actually make it to the market,” as, even given the vast range of varying applications, there is only so much demand for such devices. This paper then reviews three major variations of the gloves traditional gloves, thimbles and exoskeletons to weigh the pros and cons of each.

Table 1. Overview of commercial haptic gloves.

Device	Type	Nb fingers		Actuator	Force-feedback			Active	Weight (g)	Published price
			Wireless		Force-feedback	Tactile feed-back	Hand tracking			
Gloveone	Glove	5	yes	Electromagnetic	no	yes	yes	10	na	499 €
AvatarVR	Glove	5	yes	Electromagnetic	no	yes	yes	10	na	1100 €
Senso Glove	Glove	5	yes	Electromagnetic	no	yes	yes	5	na	599 \$ ^{1,2}
Cynteract	Glove	5	yes	Electromagnetic	yes	no	yes	5	na	Na
Maestro	Glove	5	yes	Electromagnetic	yes	yes	yes	5	590	na
GoTouchVR	Thimble	1	yes	Electromagnetic	no	yes	no	1	20	na
Tactai Touch	Thimble	1	yes	na	no	yes	no	1	29	na
CyberGrasp	Exosk.	5	no	Electromagnetic	yes	no	no	5	450	50000 \$ ³
Dexmo	Exosk.	5	yes	Electromagnetic	yes	no	no	5	320	12000 \$ ¹
HaptX	Exosk.	5	no	Pneumatic	yes	yes	yes	na	na	Na
VRgluv	Exosk.	5	yes	Electromagnetic	yes	no	no	5	na	579 \$
Sense Glove DK1	Exosk.	5	yes	Electromagnetic (brakes)	no	no	no	5	300	999 € ^{1,2}
EXOS Glove	Exosk.	5	no	Electromagnetic	yes	no	no	5	na	na
HGlove	Exosk.	3	no	DC motor	yes	no	no	9	750	30000 € ³

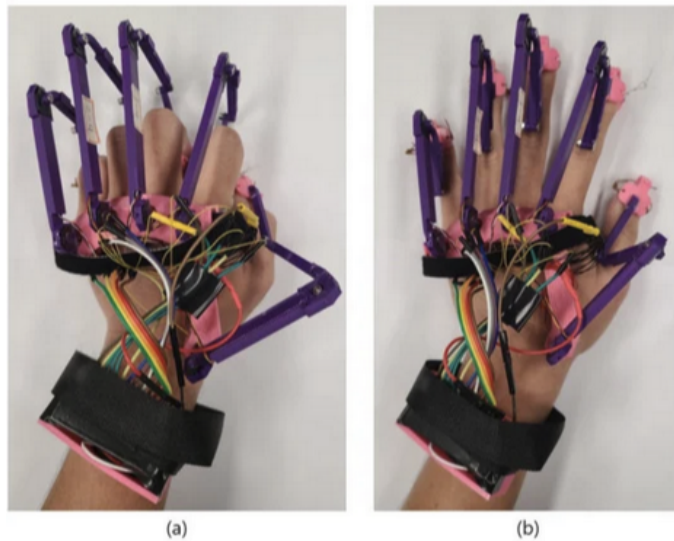
¹ Development Kit (DK)

² Two pieces (left and right)

³ Single piece manufacturing

Low-Latency Haptic Open Glove for Immersive Virtual Reality Interaction

This paper points out the need for low latency haptic gloves. It acknowledges the advances that have already been made then introduces a new method for further decreasing the latency. This new method uses rotational sensors which then senses the rotation of joints through the rigid frame attached only at the wrists and fingers. This allows for a wider range of hand types to utilise this technology. The paper also introduces a novel technique for processing the data from the sensors, further decreasing latency.



Soft robotic glove for kinesthetic haptic feedback in virtual reality environments

While many haptic gloves now try to provide haptic feedback back to the user, this mostly takes the form of vibrating motors. This is not particularly immersive as, for example, if the user tries to press a button in real life, they get more than vibrations as feedback. As a result, this paper is trying to address this by introducing “kinesthetic force feedback” using a “Soft exoskeleton and actuation.” After conducting a study, users apparently found the experience to be positive.



Haptic Glove With MR Brakes for Virtual Reality

Most haptic gloves use solid actuators and sensors (i.e. electric motors or rotational sensors) but this paper proposes to use “active liquids” such as magnetorheological (MR) fluids to actuate and sense movement instead. This system serves the dual function of actuating and sensing, removes the need for remote actuators, provides forces feedback, and test results indicate that it is apparently more efficient.

Hybrid Tele-Manipulation System Using a Sensorized 3-D-Printed Soft Robotic Gripper and a Soft Fabric-Based Haptic Glove

This paper combines a haptic glove and a separate 3D printed robotic arm. The haptic glove provides force feedback to the user, allowing them to sense the position of the robotic arm in order to better control it. The haptic glove also has attached sensors that tell the robotic arm where the hand is so that it can better mimic the hand's motions. This paper specifically examines the applications 3D printing has in this field and any benefits introducing force feedback back to the user can have in precision and control.

Works Cited

- Blake, J., and H.B. Gurocak. "Haptic Glove with MR Brakes for Virtual Reality." *IEEE/ASME Transactions on Mechatronics*, vol. 14, no. 5, Oct. 2009, pp. 606–615, <https://doi.org/10.1109/tmech.2008.2010934>. Accessed 8 Oct. 2019.
- Jadhav, Saurabh, et al. "Soft Robotic Glove for Kinesthetic Haptic Feedback in Virtual Reality Environments." *Electronic Imaging*, vol. 2017, no. 3, 29 Jan. 2017, pp. 19–24, <https://doi.org/10.2352/issn.2470-1173.2017.3.ervr-102>. Accessed 8 Oct. 2019.
- Low, Jin Huat, et al. "Hybrid Tele-Manipulation System Using a Sensorized 3-D-Printed Soft Robotic Gripper and a Soft Fabric-Based Haptic Glove." *IEEE Robotics and Automation Letters*, vol. 2, no. 2, Apr. 2017, pp. 880–887, <https://doi.org/10.1109/lra.2017.2655559>. Accessed 19 May 2019.
- Perret, J, and E. Vander Poorten. "Touching Virtual Reality: A Review of Haptic Gloves." *Review Paper: Commercial Haptic Gloves*, 25 June 2018, pp. 1–5. Accessed 30 Oct. 2024.
- Sim, Donghyun, et al. "Low-Latency Haptic Open Glove for Immersive Virtual Reality Interaction." *Sensors*, vol. 21, no. 11, 25 May 2021, p. 3682, <https://doi.org/10.3390/s21113682>.